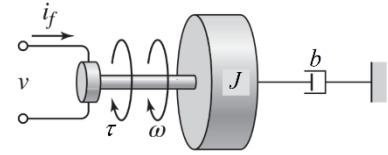


MENG 3020 – Quiz 2 Solution – Fall 2025

Question: A certain rotational system has an inertia J and viscous damping constant b . The torque $\tau(t)$ is applied by an electric motor.

The equation of motion of the mechanical subsystem is:

$$100 \frac{d\omega(t)}{dt} + 20\omega(t) = \tau(t)$$



The voltage $v(t)$ is applied to the motor. The model of the motor's field current $i_f(t)$ in amperes is:

$$0.001 \frac{di_f(t)}{dt} + 2i_f(t) = v(t)$$

The torque-current relationship is $\tau(t) = 36i_f(t)$.

a) [8 marks] Determine the transfer function models of the mechanical and electrical subsystems and complete the block diagram model of the system. Find the overall transfer function $\Omega(s)/V(s)$ of the system. What is the order of the overall system? Show your work and justify your answer.

Transfer functions of the mechanical and electrical subsystems are determined by taking Laplace transform.

Mechanical Subsystem model:

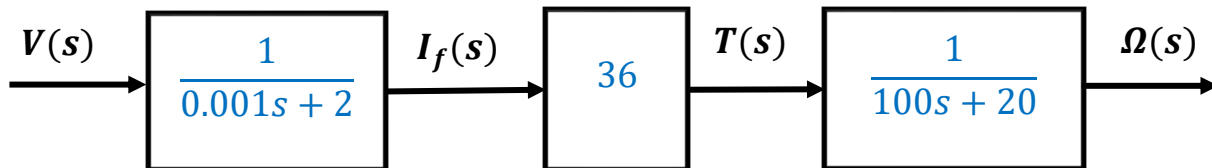
$$100 \frac{d\omega(t)}{dt} + 20\omega(t) = \tau(t) \rightarrow 100s\Omega(s) + 20\Omega(s) = T(s) \rightarrow \frac{\Omega(s)}{T(s)} = \frac{1}{100s + 20}$$

Electrical Subsystem model:

$$0.001 \frac{di_f(t)}{dt} + 2i_f(t) = v(t) \rightarrow 0.001sI_f(s) + 2I_f(s) = V(s) \rightarrow \frac{I_f(s)}{V(s)} = \frac{1}{0.001s + 2}$$

The torque-current relationship is:

$$\tau(t) = 36i_f(t) \rightarrow T(s) = 36I_f(s)$$



The overall transfer function model is:

$$\frac{\Omega(s)}{V(s)} = \left(\frac{1}{100s + 20} \right) (36) \left(\frac{1}{0.001s + 2} \right) = \frac{36}{(100s + 20)(0.001s + 2)} = \frac{36}{0.1s^2 + 200.002s + 40}$$

Since the highest order of the characteristic equation is two, the overall system is **second** order.

b) [4 marks] Determine the time-constant of the mechanical and electrical subsystems. Which one has a faster response? Which time-constant is the dominant time-constant of the overall system? Show your work and justify your answer.

Time-constant of the mechanical subsystem:

$$\frac{\Omega(s)}{T(s)} = \frac{1}{100s + 20} \quad \rightarrow \quad \tau = \frac{50}{10} = 5 \text{ sec}$$

Time-constant of the electrical subsystem:

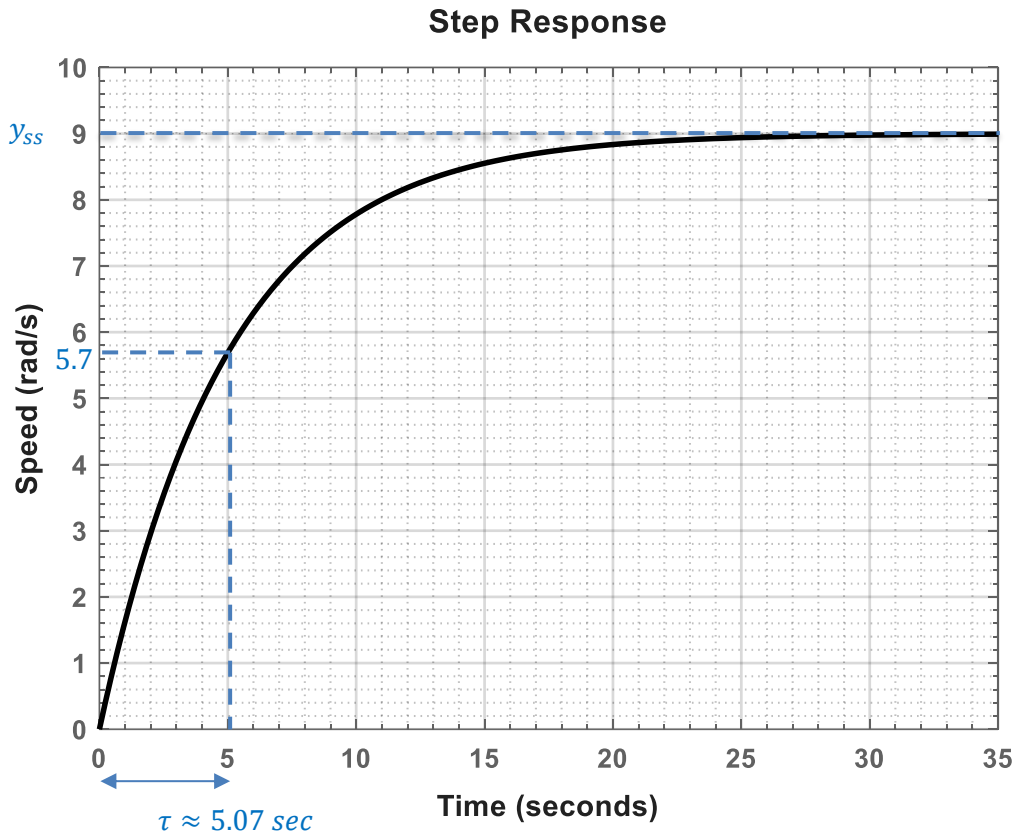
$$\frac{I_f(s)}{V(s)} = \frac{1}{0.001s + 2} \quad \rightarrow \quad \tau = \frac{0.001}{2} = 5 \times 10^{-4} \text{ sec}$$

The electrical subsystem (motor) has a smaller time-constant, which means it has a much faster response than the rotational mechanical subsystem.

The mechanical subsystem has a greater time-constant, which means it has the dominant pole. Thus, the dominant time-constant of the overall system is $\tau = 5 \text{ sec}$.

c) [8 marks] Suppose that the applied voltage is $v(t) = 10V$ step input. The following graph represents the speed profile of the system. Determine the time-constant τ and steady-state gain K of the system from the graph and estimate a first-order transfer function model for this system. Show your work and required measurements on the graph.

$$G(s) = \frac{K}{\tau s + 1}$$



Using the graph, find the steady-state gain K is

$$K = \frac{\Delta y}{\Delta u} = \frac{y_{ss} - y_0}{u_{ss} - u_0} = \frac{9 - 0}{10 - 0} = 0.9$$

The time-constant is determined using the 63.2% method

$$\Delta y \times 0.632 = 9 \times 0.632 = 5.688 \approx 5.7$$

Using the graph, find the time at the amplitude of 5.688

$$t_2 \approx 5.07 \text{ sec}$$

The time-constant is: $\tau = t_2 - t_0 = 5.07 - 0 = 5.07 \text{ sec}$

Therefore, the first-order transfer function model is:

$$G(s) = \frac{0.9}{5.07s + 1}$$